AC9M6N06 • Year 6 Mathematics

Multiplying and Dividing Decimals by Powers of 10

Track changing digit value and factor multiples of ten

We are learning to…

Students explain how each digit's place value changes under multiplication or division by 10, 100 and 1 000, then extend this to factors such as 30, 400 and 0.1-related contexts.

Represent | Reason | Calculate | Interpret | Verify

Success criteria

- I can represent or identify the concept.
- I can explain the underlying relationship.
- I can select an appropriate strategy or feature.
- I can apply it in a new context.
- I can justify and verify the response.

Curriculum focus

multiply and divide decimals by multiples of powers of 10 without a calculator, applying knowledge of place value and proficiency with multiplication facts; using estimation and rounding to check the reasonableness of answers

Track 4.275 through ×10, ×100 and ÷10

operation	result	digit-value change
4.275 × 10	**42.75**	**each digit worth 10 times as much**
4.275 × 100	**427.5**	100 times as much
4.275 ÷ 10	0.4275	one tenth as much

The decimal point is a fixed notation reference; explain changing digit values rather than saying the point physically moves.

Factor multiples of powers of ten

2.4 × 300 → 2.4 × 3 × 100 → 7.2 × 100 → 720

For 48 ÷ 0.6, scaling both numbers by 10 gives 480 ÷ 6 = 80; this preserves the quotient.

Build and annotate the model

Represent multiplying and dividing decimals by powers of 10 and label the important parts, quantities or choices.

Compare and reason

Use the application model to compare two cases, explain the relationship and identify a likely error.

Transfer and verify

Apply the idea in an unfamiliar context and use a second method, evidence source or text feature to check it.

Common mix-ups

- **Add zeros rule used with decimals:** Use place value; zeros may be placeholders, not a universal rule.
- **Decimal point said to move:** Digits change place value relative to the fixed point.
- **Factor 300 treated as 3:** Include the factor of 100.
- **Division estimate omitted:** Check quotient magnitude before accepting.

Quick check

- Calculate 4.275 × 100.
- Calculate 72.5 ÷ 10.
- Solve 2.4 × 300.
- Solve 48 ÷ 0.6.
- Explain digit-value change.

Teacher decision: continue when students explain the model and verify a new example; otherwise return to Slide 2.